TRANPLANTATION AND GRAFT REJECTION

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Learning outcome:

- By the end of this lecture students should be able to;

1- Define Organ transplantation, organ donation, graft and graft rejection.

2- List different types of organ transplantation, donors and grafts.

3- Classify transplantation rejection .

4- Identify symptoms of transplant rejection.

5- Understand stem cells and some of their clinical uses.

What is organ transplantation?

- An **organ transplantation** is a surgical operation in which a damaged or failed organ is removed and replaced by a functioning one.
- Transplantation provides the opportunity for definitive treatment of end stage organ disease.

Organ donation

Types of donors

- Living donors
- Deceased donors

Graft

- **A piece of skin, bone, organ, or other living tissue transplanted from one body, or place on a body, to another.**

Types of Grafts

- **1- Autograft**
- **2- Allograft**
- **3- Isograft**
- **4- Xenograft**

Graft rejection

- Solid organ transplantation inevitably stimulates an aggressive immune response by the recipient, unless the transplant is between monozygotic twins.
- The type and severity of the rejection response is determined by the genetic disparity between the donor and recipient, the immune status of the host and the nature of the tissue transplanted .

Graft rejection…

- The most important genetic determinant is the difference between donor and recipient HLA proteins. The extensive polymorphism of these proteins means that donor HLA antigens are almost invariably recognised as foreign by the recipient immune system, unless an active attempt has been made to minimize incompatibility.

4.20 Classification of transplant rejection

Type	Time	Pathological findings	Mechanism	Treatment
Hyperacute rejection	Minutes to hours	Thrombosis, necrosis	Pre-formed antibody to donor antigens results in complement activation (type II hypersensitivity)	None – irreversible graft loss
Acute cellular rejection	5–30 days	Cellular infiltration	$CD4^+$ and $CD8^+$ T cells (type IV hypersensitivity)	Increase immunosuppression
Acute vascular rejection	5–30 days	Vasculitis	Antibody and complement activation	Increase immunosuppression
Chronic allograft failure	> 30 days	Fibrosis, scarring	Immune and non-immune mechanisms	Minimise drug toxicity, control hypertension and hyperlipidaemia

Hyperacute rejection

- Results in rapid and irreversible destruction of the graft .
- It is mediated by pre-existing recipient antibodies against donor HLA antigens, which arise as a result of previous exposure through transplantation, blood transfusion or pregnancy.
- It is very rarely seen in clinical practice, as the use of screening for anti-HLA antibodies and pre-transplant cross-matching ensures the prior identification of such recipient–donor incompatibility.

Acute cellular rejection

- is the most common form of graft rejection.
- It is mediated by activated T lymphocytes and results in deterioration in graft function.
- If allowed to progress, it may cause fever, pain and tenderness over the graft. It is usually amenable to increased immunosuppressive therapy.

Acute vascular rejection

- Is mediated by antibody formed de novo after transplantation.
- It is more curtailed than the hyperacute response because of the use of intercurrent immunosuppression but it is also associated with reduced graft survival.
- Aggressive immunosuppressive therapy is indicated and physical removal of antibody through plasmapheresis may be indicated in severe causes.
- Not all post-transplant anti-donor antibodies cause graft damage; their consequences are determined by specificity and ability to trigger other immune components, such as the complement cascade.

Chronic allograft failure

- Also known as chronic rejection.
- is a major cause of graft loss.
- It is associated with proliferation of transplant vascular smooth muscle, interstitial Fibrosis and scarring.
- The pathogenesis is poorly understood but contributing factors include immunological damage caused by subacute rejection, hypertension, hyperlipidaemia and chronic drug toxicity.

Symptoms of transplant rejection

- Pain at the site of transplant
- Feeling ill
- Flu like symptoms
- Fever
- Weight change
- Swelling
- Decreased urine output

Stem cells

- What are stem cells?
- Types of stem cells?
- Stem cell transplantation
- Haematopoietic stem cell transplantation
- ? Bone marrow transplantation

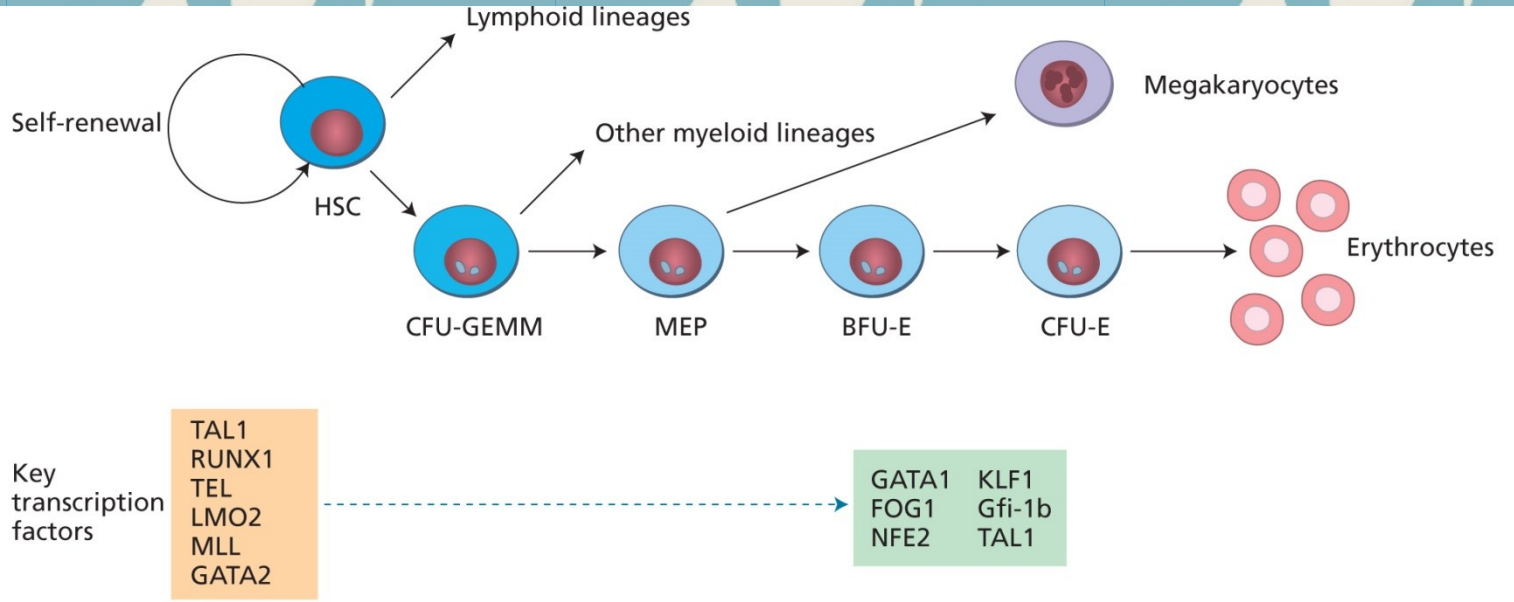


Figure 2.1 Summary of some steps in self-renewal, lineage specification and differentiation of haemopoietic stem cells to red cells. Some of the key transcription factors involved in this process are summarized beneath the diagram. HSC, haemopoietic stem cell; CFU-GEMM, colony-forming unit - granulocyte erythrocyte monocyte megakaryocyte; MEP, megakaryocyte-erythroid progenitor; BFU-E, burst-forming unit - erythroid; CFU-E, colony-forming unit - erythroid; TAL1, T-cell acute lymphoblastic leukaemia 1; TEL, translocation Ets leukaemia; LMO2, LIM domain only 2; MLL, mixed lineage leukaemia; GATA1, GATA-binding factor 1; GATA2, GATA-binding factor 2; FOG, friend of GATA; NFE2, nuclear factor erythroid-derived 2; KLF1, Kruppel-like factor 1 (erythroid); GFi-1b, growth-factor independent 1b.

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Types of HSCTs

- Autologus
- Allogeniec

ANY QUESTIONS?